



Built-in anti-cheat in Windows

Microsoft secretly added an anti-cheat mechanism called TruePlay into Windows through the Fall Creators Update. <http://dgit.in/MSAntiCheat>



Death of Denuvo?

Once the best-in-class DRM mechanism for video games, Denuvo is now being cracked in under 24 hours of release. <http://dgit.in/denuvodead>

Radeon RX Vega 64 vs NVIDIA GTX 1080

Finally, a battle of flagship GPUs after many years of waiting.

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MD's re-entry into the CPU and GPU space have been welcome with open arms. The Zen-based Ryzen CPUs have ignited the flames of competition once again in the CPU space between Intel and AMD. And on the side, the Radeon Technologies Group (RTG) has done the same with their Radeon RX Vega GPUs. While RTG doesn't have as many aftermarket SKUs as NVIDIA does, we still have more than enough. And both companies faced stock issues at the onset of their respective launches, however, the controversy around RX Vega was a lot more pronounced than what the 10-series flagships from NVIDIA managed to evoke. However, it's been more than a year since NVIDIA's 10-series came out and barely two months since the Vega GPUs came out. Prima facie, it would appear Vega didn't up the ante. However, matching performance is a significant step in the right direction as well. Let's take a look at the GPUs we'll be comparing today.

The two GPUs have vastly different microarchitectures and we can't simply compare them against each other with respect to the individual components at the silicon level. That would be sacrilege, so we're simply stating elements of the architecture that are the most similar to provide a perspective from the get go. The Vega 64 has 64 Compute Units that hold a total of 4096 stream processors and the GTX 1080 has 20 Streaming Multiprocessors which hold 2560 CUDA cores. These are the essential units of computation within GPUs. The Vega 64 GPU has significantly higher memory bus width (2048-bit) while the GTX 1080 uses a 256-bit bus since it has fewer cores that need to simultaneously communicate. AMD's GPU is built on Global Foundries' 14nm process node while NVIDIA uses the TSMC 16nm node. It would appear AMD has the more power-efficient node, however, they have more stream processors overall, so all the saving gets eaten up quickly. For this comparison, we're looking at the reference SKUs from both competitors.